

Adarsh Kumar Singh

AI & Systems Engineer | GenAI, Backend Architecture & Embedded Systems

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WHO I AM

I am an Electronics and Communication engineer who fell in love with Artificial Intelligence, Database Systems, and Backend Architecture. I build things that bridge the gap between physical hardware and advanced AI—whether that means architecting an autonomous drone network for search-and-rescue, or deploying multi-agent swarms that autonomously heal CI/CD pipelines. I hold multiple Oracle Professional Certifications in GenAI and Data Science, and I thrive in environments where I can build resilient systems from the ground up. Oh, and I'm currently teaching myself Japanese and German.

CORE TECHNICAL ARSENAL

- **AI & Generative AI:** Multi-Agent Swarms (ReAct loops), Retrieval-Augmented Generation (RAG), Tool Calling, ChromaDB Vector Search, Semantic Chunking, Reranking, Prompt Guardrails, LLM-as-a-Judge.
 - **Backend & Systems:** Python, Bash, FastAPI, Linux, REST APIs, SSE Streaming, Pytest, Sandboxed Testing, System Engineering.
 - **Database & Cloud:** Oracle Cloud Infrastructure (OCI), Oracle Database Design, Advanced SQL, PostgreSQL, Data Warehousing.
 - **Machine Learning:** Scikit-learn, Pandas, NumPy, Predictive Modeling, Exploratory Data Analysis.
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ENGINEERING PROJECTS

Autonomous Self-Healing DevOps Swarm

- **Domain:** Agentic Workflows, DevOps Automation

- **What I Built:** A capstone multi-agent orchestration framework that intercepts CI/CD pipeline failures without human intervention.
- **Key Achievements:** Built agents that automatically diagnose root causes, write code patches, verify fixes via sandboxed testing environments, and autonomously open Pull Requests with the verified fixes.

Enterprise Production Agent & Eval Dashboard

- **Domain:** LLM Guardrails & Production Reliability
- **What I Built:** A hardened compliance AI agent focused entirely on safety, reliability, and continuous evaluation.
- **Key Achievements:** Engineered strict Input/Output guardrails blocking prompt injection and out-of-scope queries. Implemented persistent memory, structured logging, and a real-time LLM-as-a-Judge Evaluation Dashboard with automatic Model Failover routing (switching from 70B to 8B models automatically during failures).

PersonaDoc — Production RAG System

- **Domain:** Document Intelligence, Vector Search
- **What I Built:** A robust document intelligence backend capable of ingesting any PDF, chunking it semantically, and providing exact source citations.
- **Key Achievements:** Integrated ChromaDB vector search, semantic chunking, and advanced reranking. Built comprehensive hallucination control logic to ensure perfectly accurate answers.

AI Code Review Service

- **Domain:** CI/CD Integration, Applied LLMs
- **What I Built:** A service that takes any GitHub PR URL and connects directly to the real GitHub diffs to perform structured code reviews.
- **Key Achievements:** Engineered the LLM to output structured JSON identifying bugs, security vulnerabilities, assigning severity ratings, and issuing approval decisions.

Search-and-Rescue Drone System *(4th-Year Major Project | Oct 2025 - May 2026)*

- **Domain:** Embedded Systems, IoT, Real-Time Databases

- **What I Built:** Designed and architected an autonomous drone system aimed at revolutionizing emergency response times in critical rescue operations.
 - **Key Achievements:** Engineered the database schema to handle massive influxes of real-time sensor data from multiple drones. Integrated hardware controllers, communication protocols, and back-end infrastructure into a cohesive system.
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EDUCATION

B.Tech in Electronics and Communication Engineering

Dr. Abdul Kalam Technical University (AKGEC) | Ghaziabad, Uttar Pradesh

Expected Graduation: July 2026 | CGPA: 8.2

CERTIFICATIONS & CONTINUOUS LEARNING

- **Oracle GenAI Certified Professional** (2025)
 - **Oracle Data Science Professional Certification** (2025)
 - **Oracle Certified Data Science Professional** (Database 23 AI)
 - **IoT and Industrial Automation Certification** (2025)
 - **EC Council Code-Red Certifications** (Cybersecurity & Systems)
 - **GFG Machine Learning and Data Science Certification** (*Ongoing*)
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ADDITIONAL DIMENSIONS

- **Fast Learner:** I adapt rapidly to new client requirements, business domains, and tech stacks. I am ready to deploy code and contribute to a team's success from day one.
- **Global Mindset:** Currently expanding my communication capabilities by actively learning **Japanese** and **German**.